

	Work Instruction	Doc. Revision	00
	AXI V810 Trigger Board Power Supply (2140-0039) Upgrade Procedure	Effective Date	12 Sep 2017

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	12 Sep 2017	First Release	ONG KEAN SHENG

	Work Instruction	Doc. Revision	00
	AXI V810 Trigger Board Power Supply (2140-0039) Upgrade Procedure	Effective Date	12 Sep 2017

Content

Trigger Board Power Supply Removal.....	3
Trigger Board Power Supply Replacement.....	9

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 Trigger Board Power Supply (2140-0039) Upgrade Procedure	Effective Date	12 Sep 2017

Trigger Board Power Supply Removal

1. Turn Off X Ray Source
2. Power Off entire machine
3. Turn Off Trigger Board Power Switch (PDU Rear Area). Unplug the power cord as well as below
Figure 1



Figure 1

4. Unscrew and remove TOP, TRIGGER BOX (P/N, 4XACM-007).

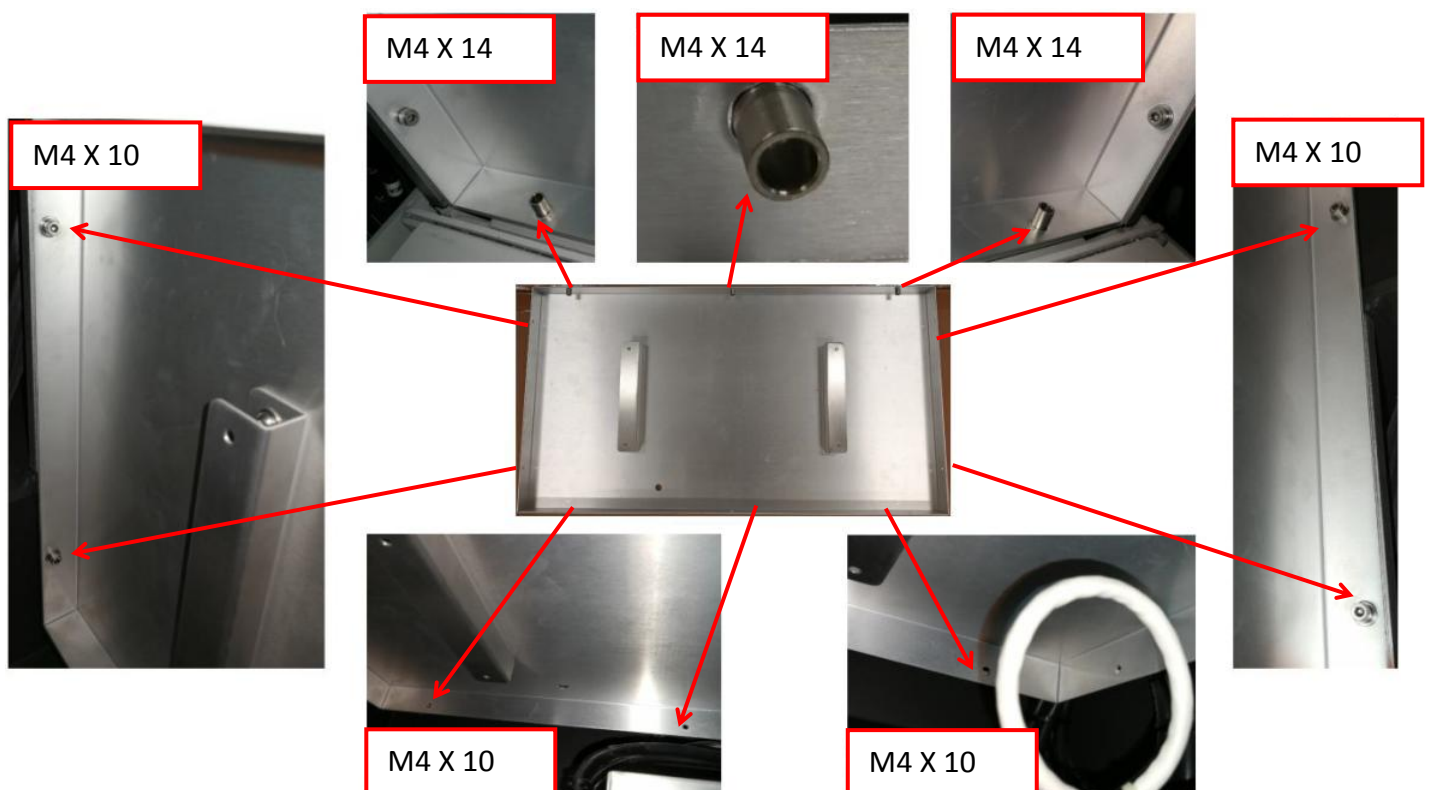


Figure 2 Trigger board Tray Cover removal

	Work Instruction	Doc. Revision	00
	AXI V810 Trigger Board Power Supply (2140-0039) Upgrade Procedure	Effective Date	12 Sep 2017

- Disconnect 14 x Fire Wires from Embedded System Support MISC Board-Trigger Board 1 Input Channel 12 Output Channels ports (P/N, 2210-0016) as Figure 3.



Figure 3 Trigger board

- Disconnect the power cable from J424v connector as Figure 4.

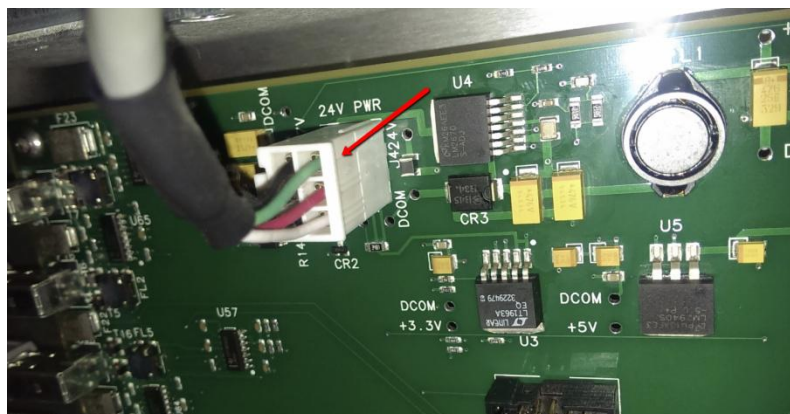


Figure 4 Trigger board power removal

- Use a short slotted screwdriver unscrew and disconnect the CABLE, TRIGGER IO, DBN 15 (P/N, 4XASC-A039) cable from Trigger Board as Figure 5.

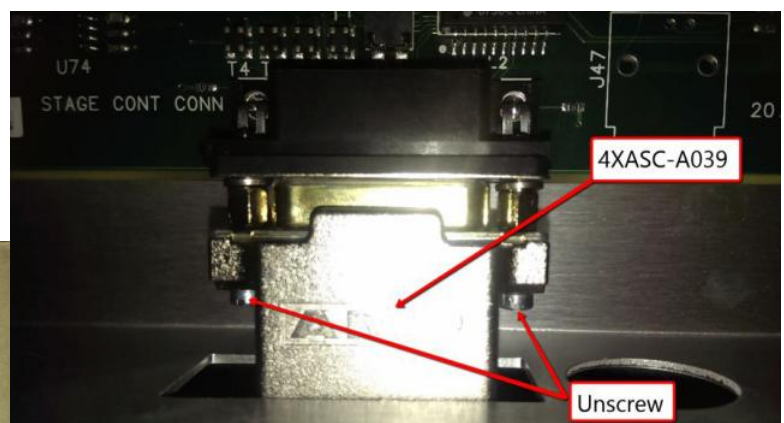


Figure 5 Connector removal

	Work Instruction	Doc. Revision	00
	AXI V810 Trigger Board Power Supply (2140-0039) Upgrade Procedure	Effective Date	12 Sep 2017

8. Remove Life, Neutral & Ground Cable From Trigger Board Power Switch as below Figure 6

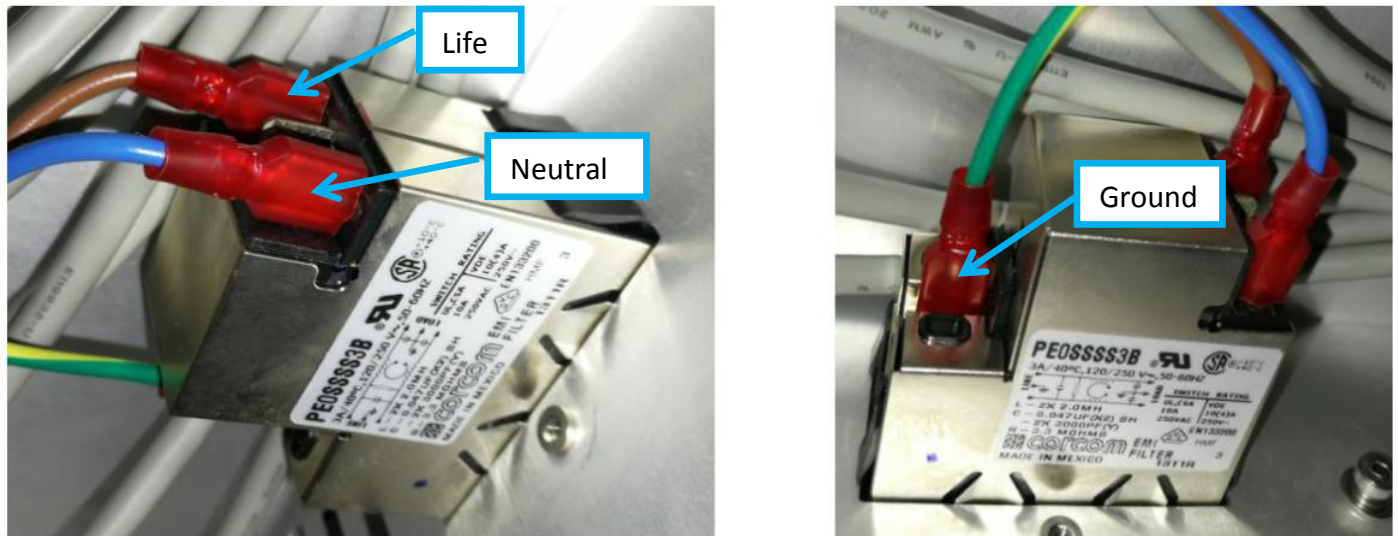


Figure 6 Power cable removal

9. Gently press on both side of Power Switch simultaneously by using Flat Head Screw Driver & remove accordingly.

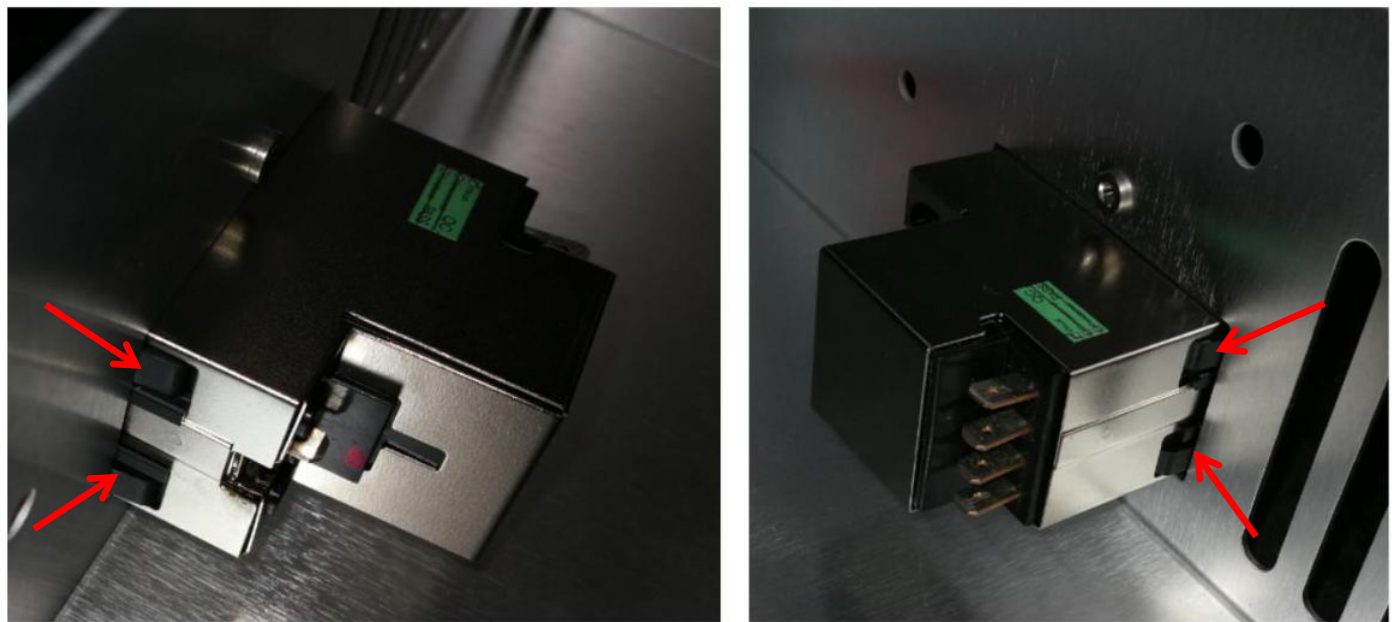


Figure 7 Power switch removal

	Work Instruction	Doc. Revision	00
	AXI V810 Trigger Board Power Supply (2140-0039) Upgrade Procedure	Effective Date	12 Sep 2017

10. Loosen 5 x screws at Trigger Board Tray as below Figure 8

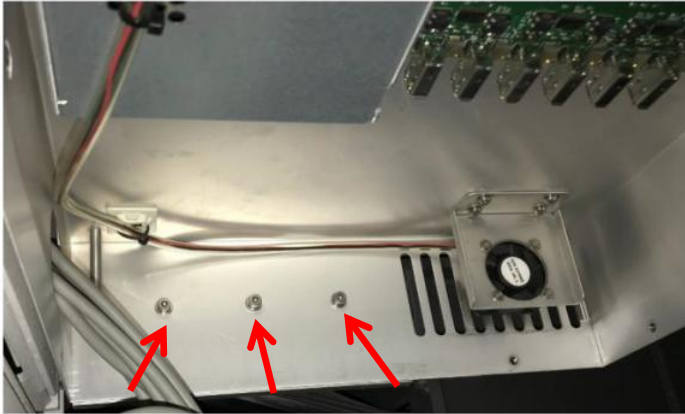


Figure 8 Trigger Board Tray removal

11. Gently remove 14 x Fire Wire through both tray opening as below Figure 9

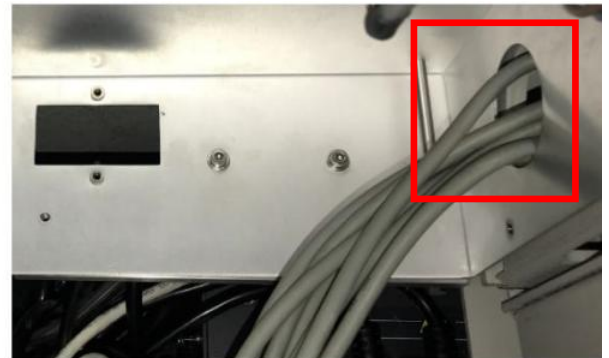
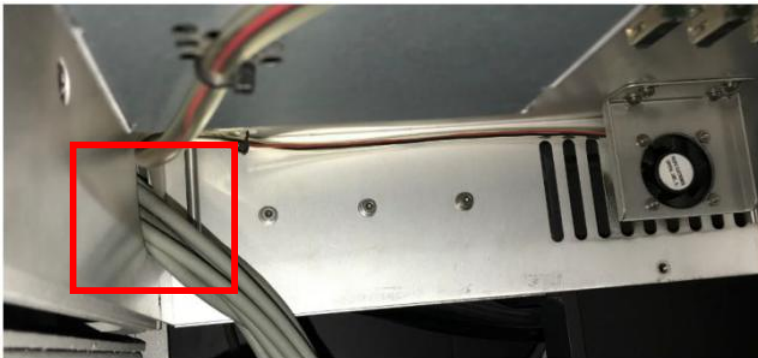


Figure 9 Fire Wire removal

12. Gently remove entire Trigger Board Tray to visible location



Figure 10 Trigger Board Tray

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 Trigger Board Power Supply (2140-0039) Upgrade Procedure	Effective Date	12 Sep 2017

13. Flip the Trigger Board Tray. Loosen 4 x Screws to remove Trigger Board Power Supply as below Figure 11.

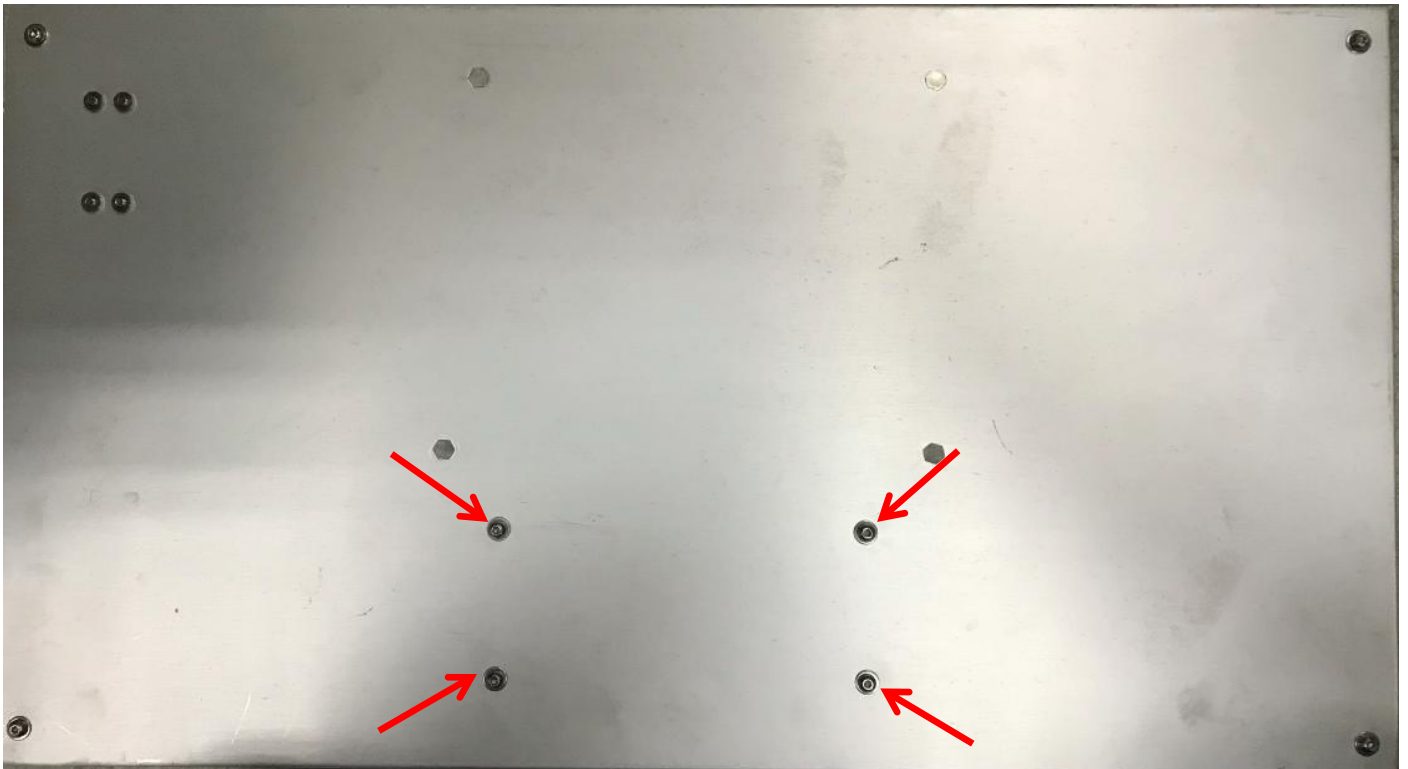


Figure 11 Trigger Board Power Supply removal

14. Remove Cable Ties at Trigger Board Power Supply Convection Casing as below Figure 12

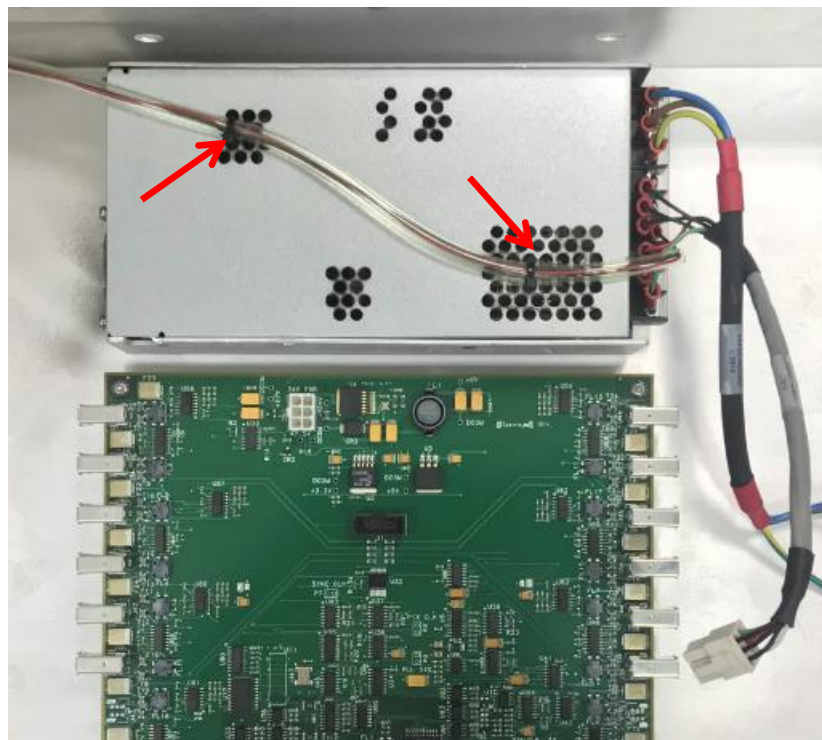


Figure 12 Cable Tie removal

	Work Instruction	Doc. Revision	00
	AXI V810 Trigger Board Power Supply (2140-0039) Upgrade Procedure	Effective Date	12 Sep 2017

15. Remove Ventilation Fan Power Cable from Trigger Board Power Supply as below Figure 13.

Remarks : May identify through Ventilation Fan Routing Or Tubing Protector.

Perform Marking upon cable removal

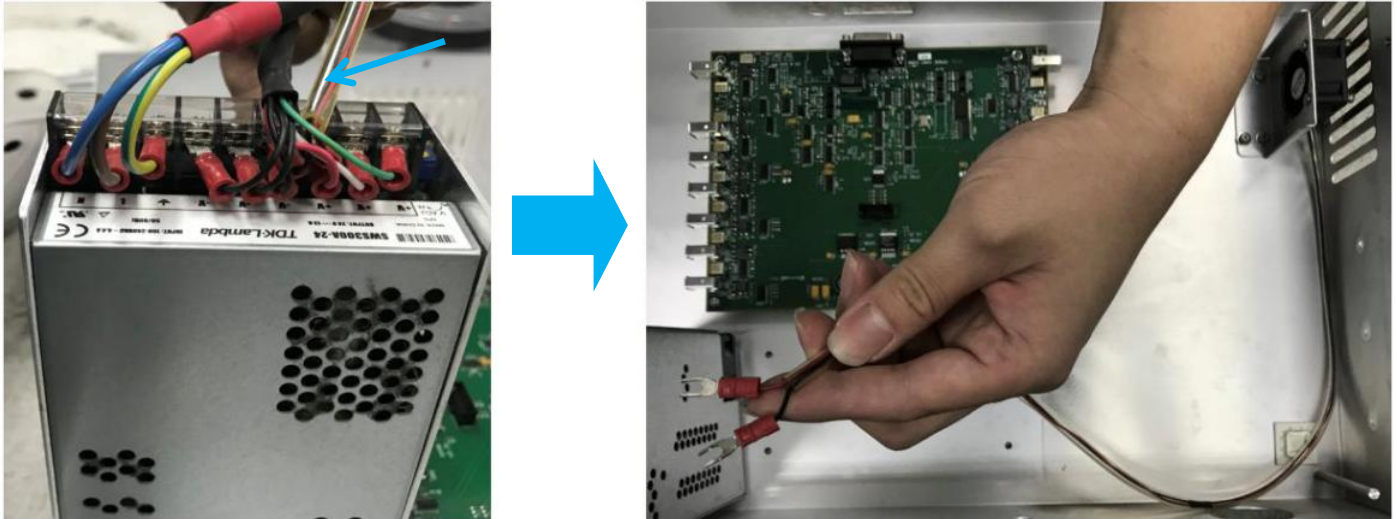


Figure 13 Ventilation Fan Power Cable Removal

16. Remove Trigger Board End Power Cable from Power Supply as below Figure 14.

Remarks : Perform Marking upon cable removal

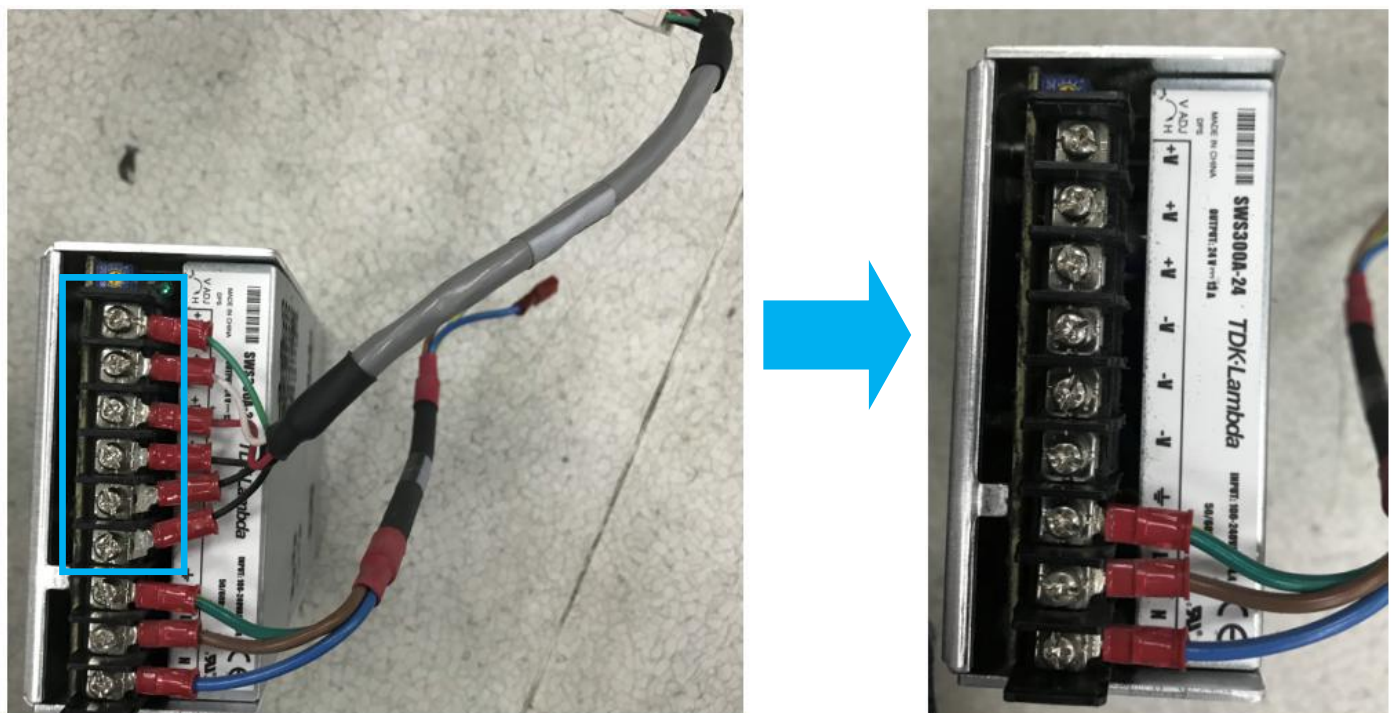


Figure 14 Trigger Board End Power Cable removal

	Work Instruction	Doc. Revision	00
	AXI V810 Trigger Board Power Supply (2140-0039) Upgrade Procedure	Effective Date	12 Sep 2017

17. Remove Power Switch Cable from Power Supply as below Figure 15

Remarks : Perform Marking upon Cable Removal



Figure 15 Power Switch Cable Removal

Trigger Board Power Supply Replacement

1. Prepare below parts :

AC/DC CONVERTER 24V 300W (P/N, 2140-0039) - Figure 16

Trigger Board Adapter Kit (P/N, 4XASPP-A052) - Figure 17



Figure 16 Trigger Board Power Supply

	Work Instruction	Doc. Revision	00
	AXI V810 Trigger Board Power Supply (2140-0039) Upgrade Procedure	Effective Date	12 Sep 2017



PN : 2720-0003
M3 X 8 Screw
Qty : 2 pcs



PN : 2720-0008
M4 X 16 Screw
Qty : 4 pcs



PN : 2730-0020
M4 X 10 Screw
Qty : 4 pcs



PN : 2760-0008
M4 Spring Lock
Qty : 4 pcs



PN : 2760-0004
M4 Round
Washer
Qty : 4 pcs

Figure 17 Trigger Board Adapter Kit

- Connect Power Switch Cable to Trigger Board Power Supply by refer to below Table 1.

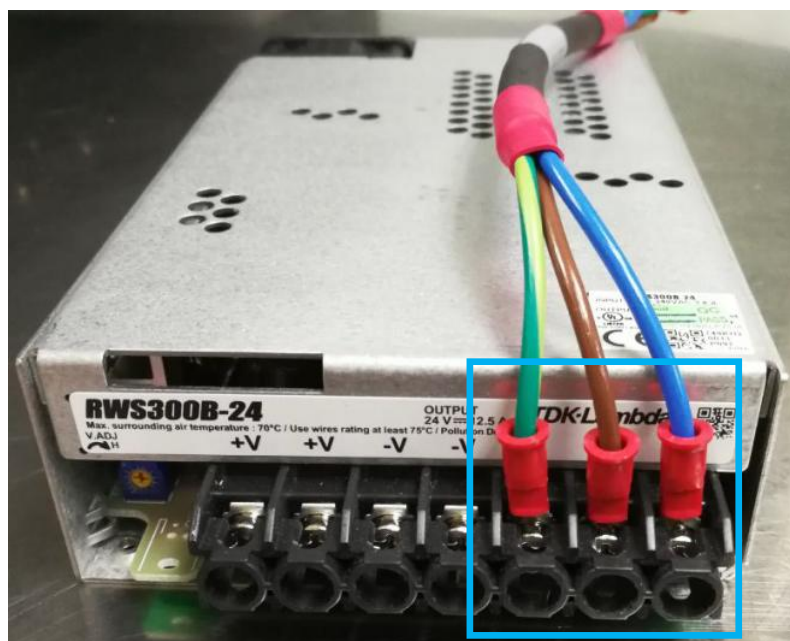


Figure 18 Power Switch Cable Connection

Power Switch Cable	Ground (Green)	Life (Brown)	Neutral (Blue)
Trigger Board Power Supply		L	N

Table 1

3. Connect Trigger Board End Power Cable to Power Supply by refer to Table 2

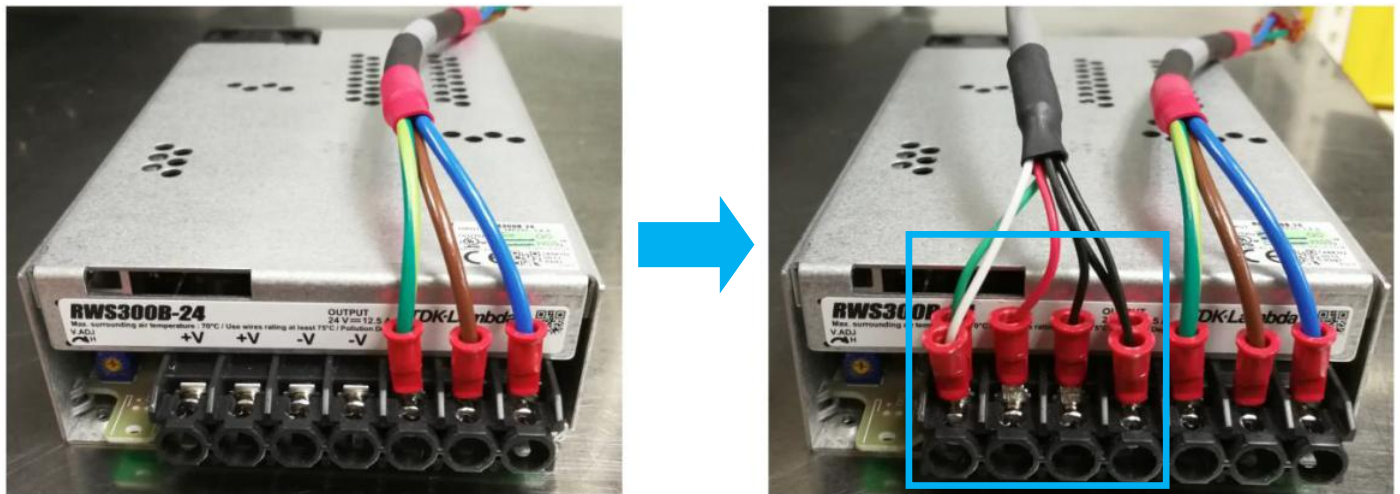


Figure 19 Trigger Board End Power Cable Connection

Trigger Board End Power Cable Colour	Green & White	Red	Black	Black & Black
Trigger Board Power Supply	+V	+V	-V	-V

Table 2

4. Connect Ventilation Fan Power Cable to Trigger Board Power Supply by refer to Table 3 (Red Colour Highlight)

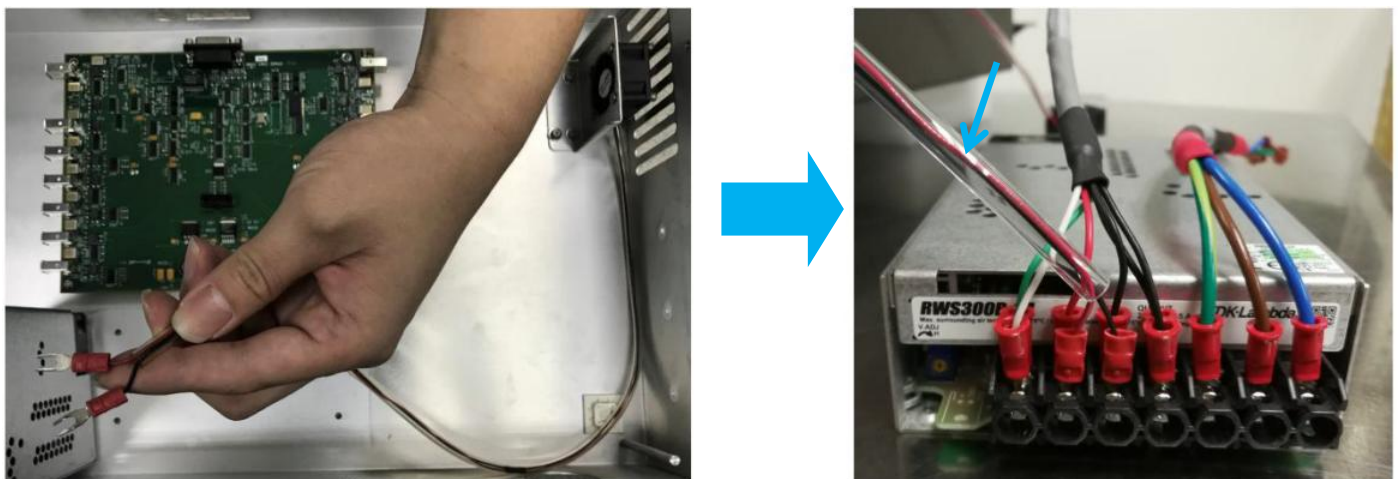


Figure 20 Ventilation Fan Power Cable Connection

Ventilation Fan Power Cable	NA	Red	Black	NA
Trigger Board Power Supply	+V	+V	-V	-V

Table 3

	Work Instruction	Doc. Revision	00
	AXI V810 Trigger Board Power Supply (2140-0039) Upgrade Procedure	Effective Date	12 Sep 2017

5. Flip the Power Supply Adapter. While attach New Trigger Board Supply to Adapter (4XAMI-099), tighten below fastening kit as below Figure 21

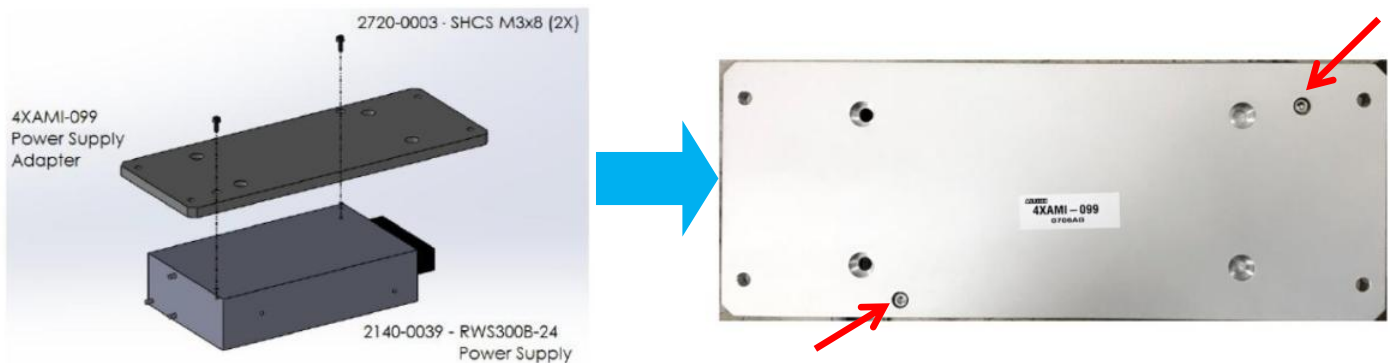


Figure 21 Trigger Board Adapter Assembly

6. Flip the Trigger Board Tray. While attach another Adapter (4XAMIS-N101) into Trigger Board Tray, tighten below fastening kit (M4 X10 Button Head Cap Screws, M4 Spring Lock & M4 Flat Washer) as below Figure 22

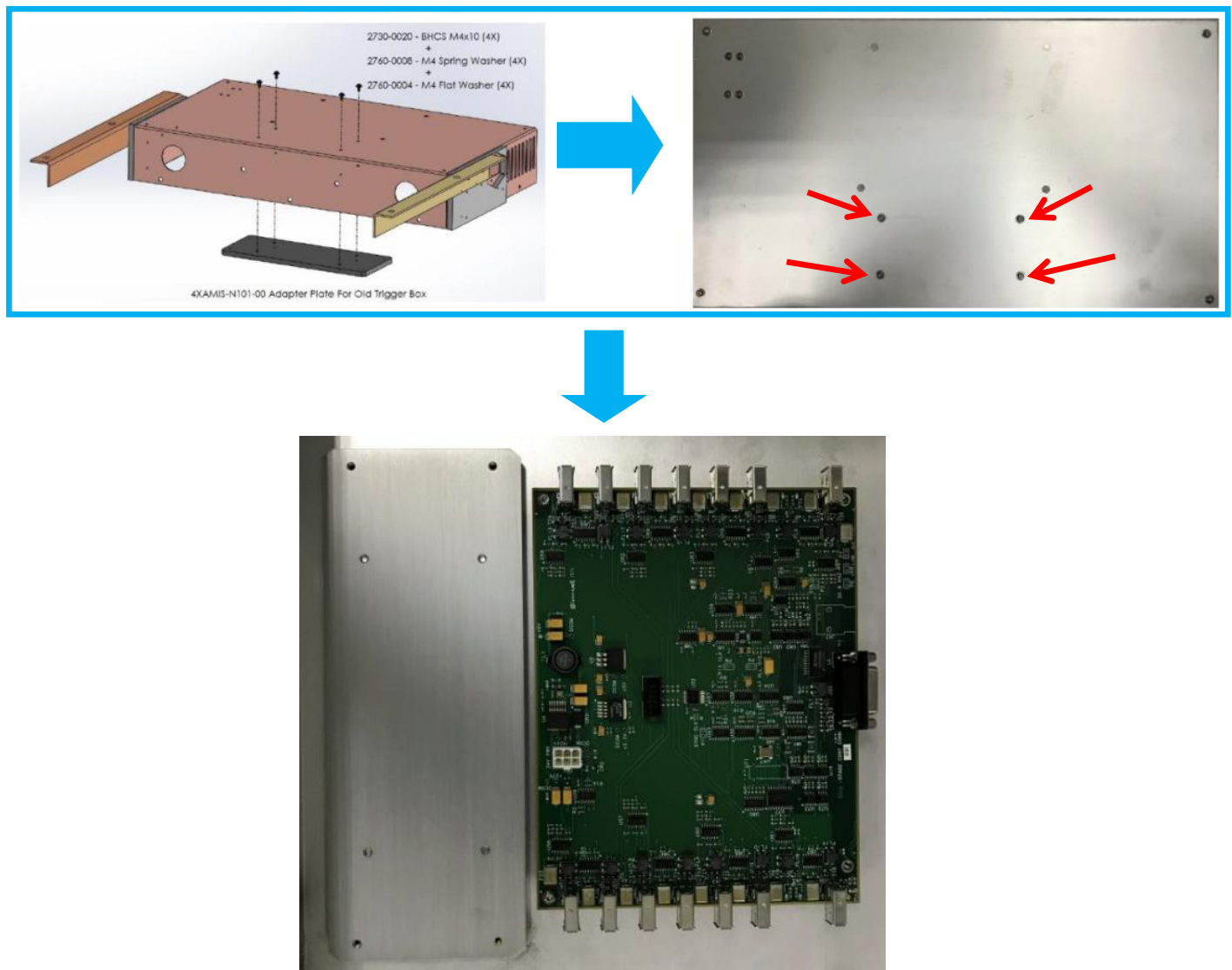


Figure 22 Adapter Assembly

	Work Instruction	Doc. Revision	00
	AXI V810 Trigger Board Power Supply (2140-0039) Upgrade Procedure	Effective Date	12 Sep 2017

7. Assemble Trigger Board Power Supply & Adapter at Trigger Board Tray as below Figure 23. Tighten 4 pcs M4 x 16 Screws & M4 Spring Locks

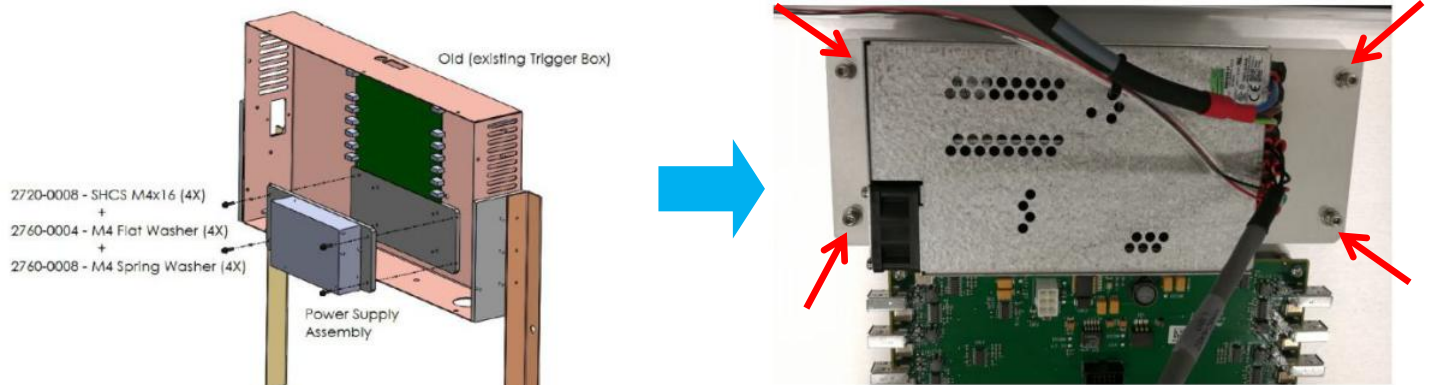


Figure 23 Trigger Board Power Supply & Adapter Assembly

8. Secure Ventilation Fan Power Cable with Cable Ties at Trigger Board Power Supply Convection Casing as below Figure 24



Figure 24

	Work Instruction	Doc. Revision	00
	AXI V810 Trigger Board Power Supply (2140-0039) Upgrade Procedure	Effective Date	12 Sep 2017

9. Assemble entire Trigger Board Tray to origin location. Tighten 5 x screws at Trigger Board Tray as below Figure 25

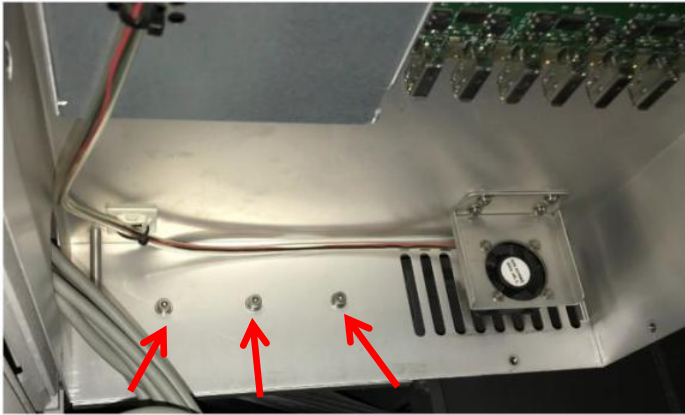


Figure 25 Trigger Board Tray assembly

10. Gently pull 14 x Fire Wire into both tray opening as below Figure 26

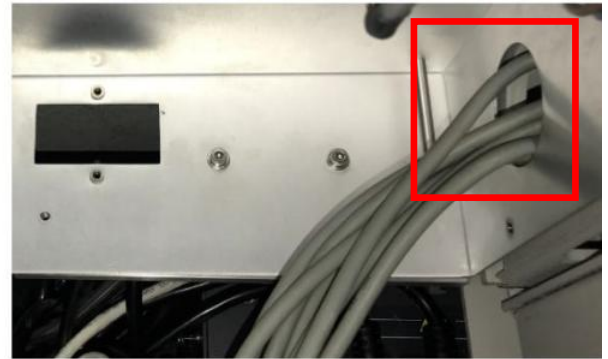
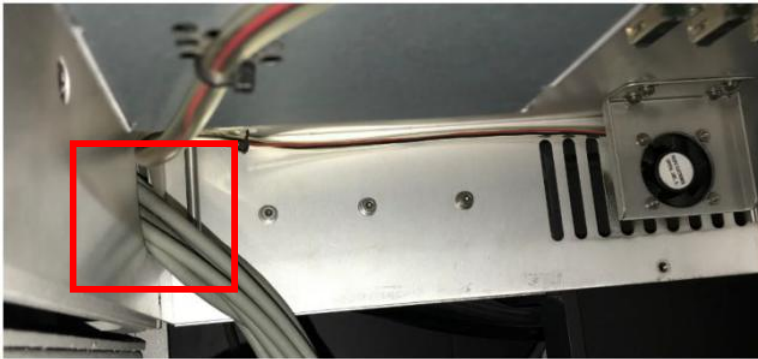


Figure 26 Retrieve 14 X Fire Wire

11. Gently connect the Power Switch by refer to below orientation to Trigger Board Tray. Connect the Power Cord as well as Figure 27

Remarks : May assemble from external



Figure 27 Power Switch assembly

	Work Instruction	Doc. Revision	00
	AXI V810 Trigger Board Power Supply (2140-0039) Upgrade Procedure	Effective Date	12 Sep 2017

12. Connect Power Switch Ground Cable into Power Switch Ground Slot as below Figure 28



Figure 28 Power Switch Ground Cable connection

13. Connect Power Switch Life & Neutral Cable into Power Switch by refer to below orientation. May refer to Figure 29

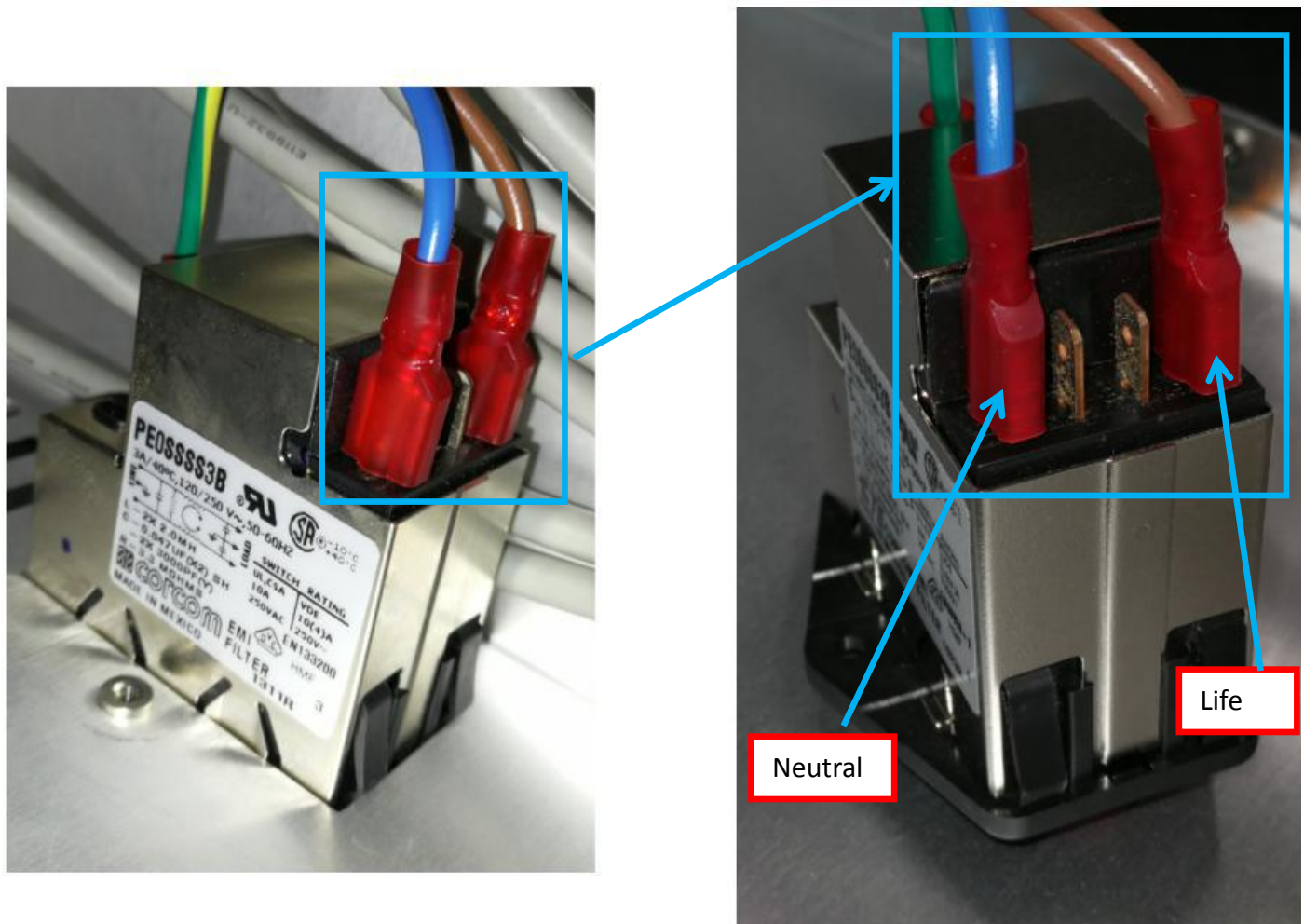


Figure 29 Power Switch Power Cable connection

	Work Instruction	Doc. Revision	00
	AXI V810 Trigger Board Power Supply (2140-0039) Upgrade Procedure	Effective Date	12 Sep 2017

14. Connect the CABLE, TRIGGER IO, DBN 15 (P/N, 4XASC-A039) cable into Trigger Board as Figure 30

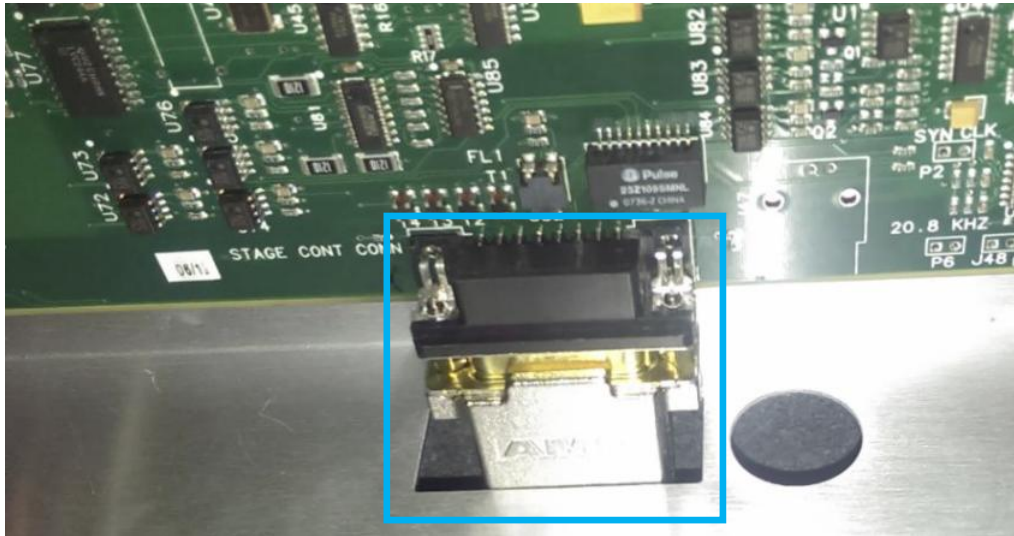


Figure 30 Trigger Board IO Cable connection

15. Connect Trigger Board Power Cable End Connector into Trigger Board (J424v Connector Slot) as Figure 31

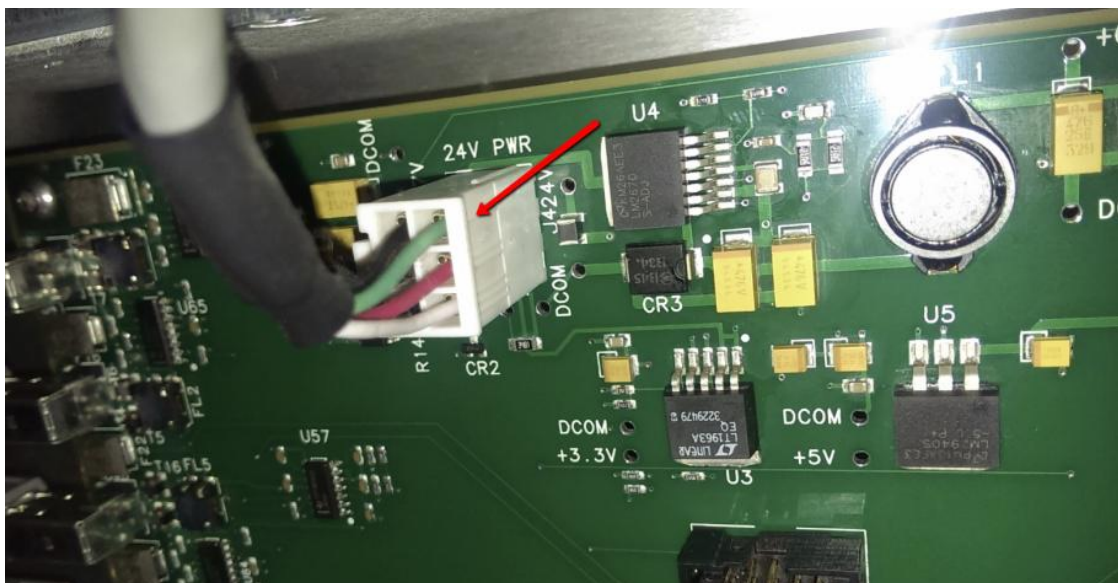


Figure 31 Trigger Board Power Connector

	Work Instruction	Doc. Revision	00
	AXI V810 Trigger Board Power Supply (2140-0039) Upgrade Procedure	Effective Date	12 Sep 2017

16. Connect 14 x Fire Wires into Trigger Board Fire Wire Port as below Figure 32

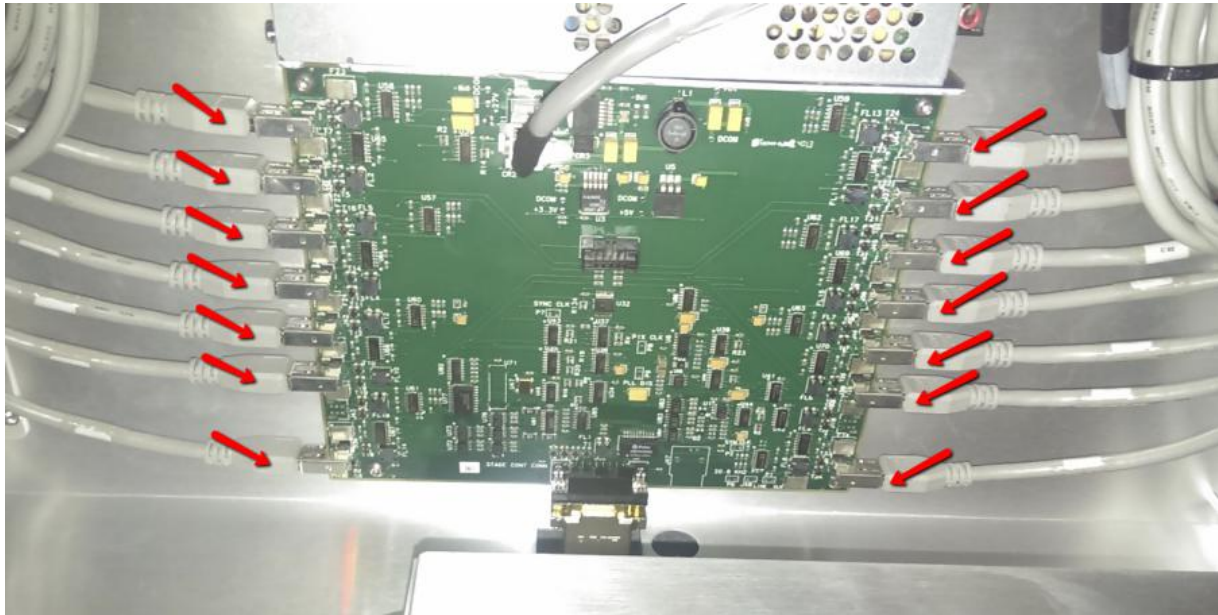


Figure 32 Fire Wire connection

17. Assemble the TOP, TRIGGER BOX (P/N, 4XACM-007) back to origin position. Tighten 9 x Screws.

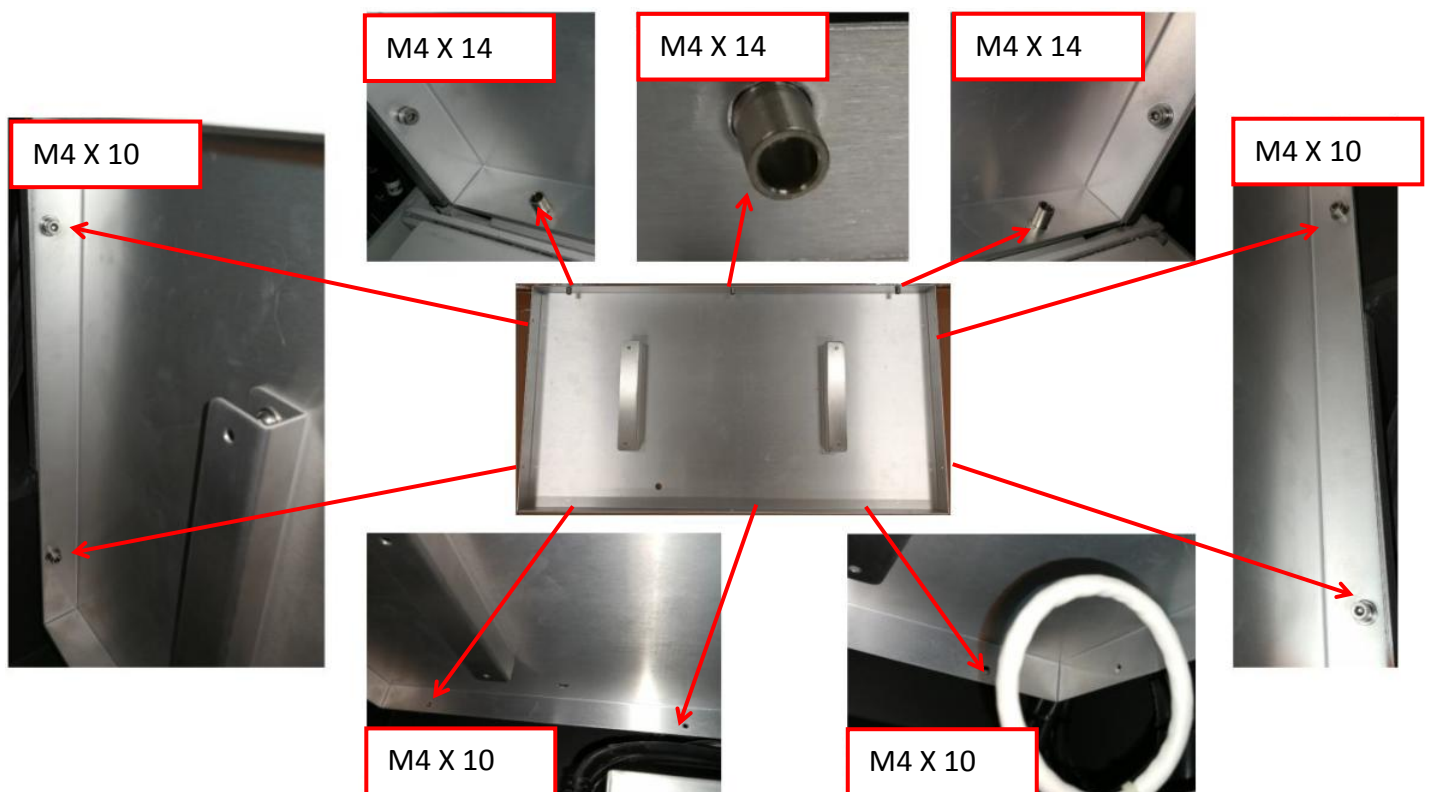


Figure 33 Trigger Board Tray Cover assembly

18. Power-up machine as usual.

19. Run 1 loop of C D & A